AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 12 · STUDENT LESSON

Speech and Audio Systems

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

AI-2 Week 26: inspect intermediate stages to locate errors.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of speech pipelines and multimodal evidence. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Examples and Rubrics. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 12: Speech and Audio Systems

Speech interaction includes **capture**, recognition, interpretation, and response stages.



CREATOR MISSION

Trace errors from noisy audio to wrong action.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall Image generation and provenance. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

This week’s end goal

By the end, I can separate **speech recognition**, calculation and speech **output** when diagnosing an error.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Speech and Audio Systems case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use speech and audio systems to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

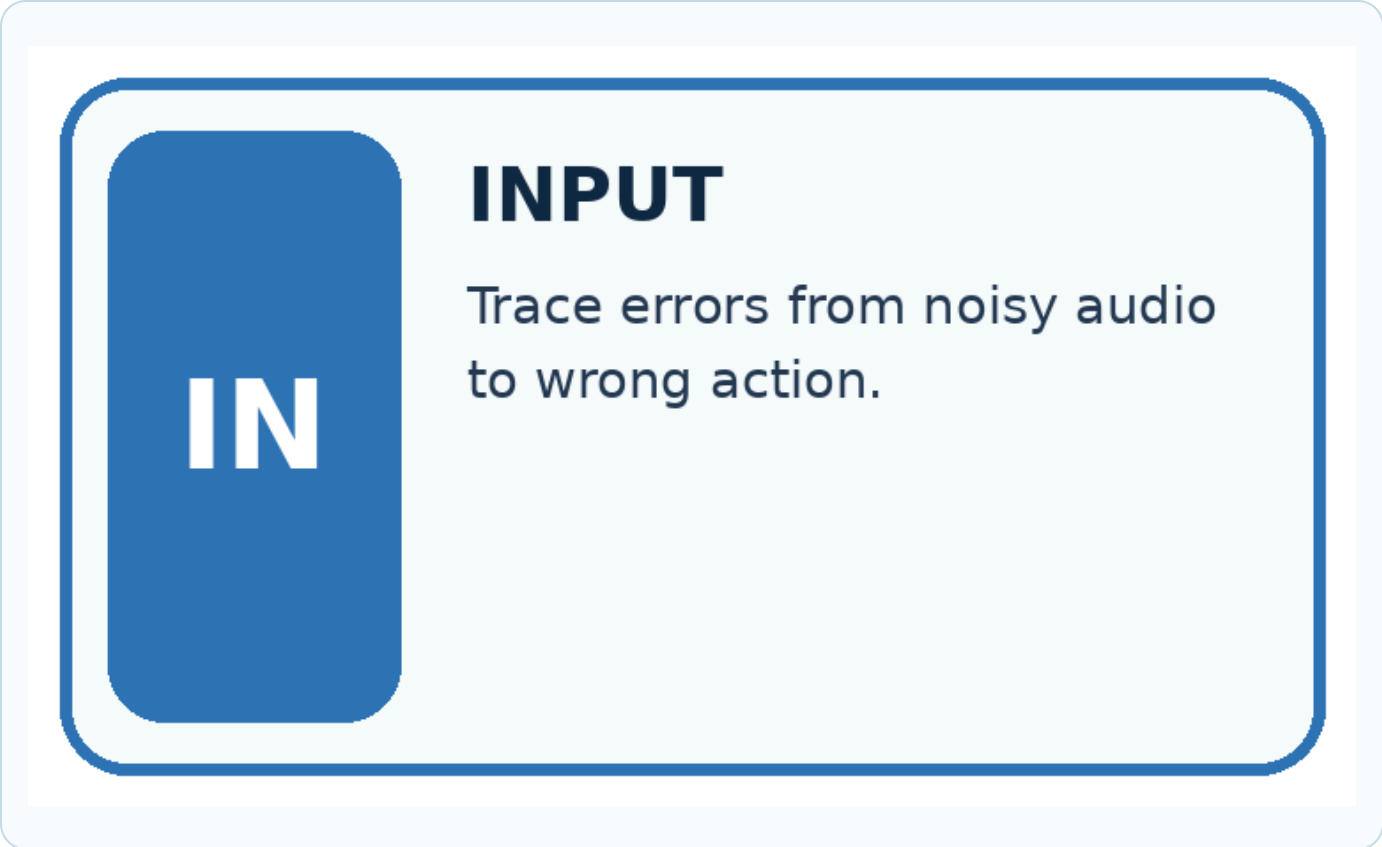
See how it works

> Follow the mechanism for Speech and Audio Systems. Connect each labelled input, deciding step and result.

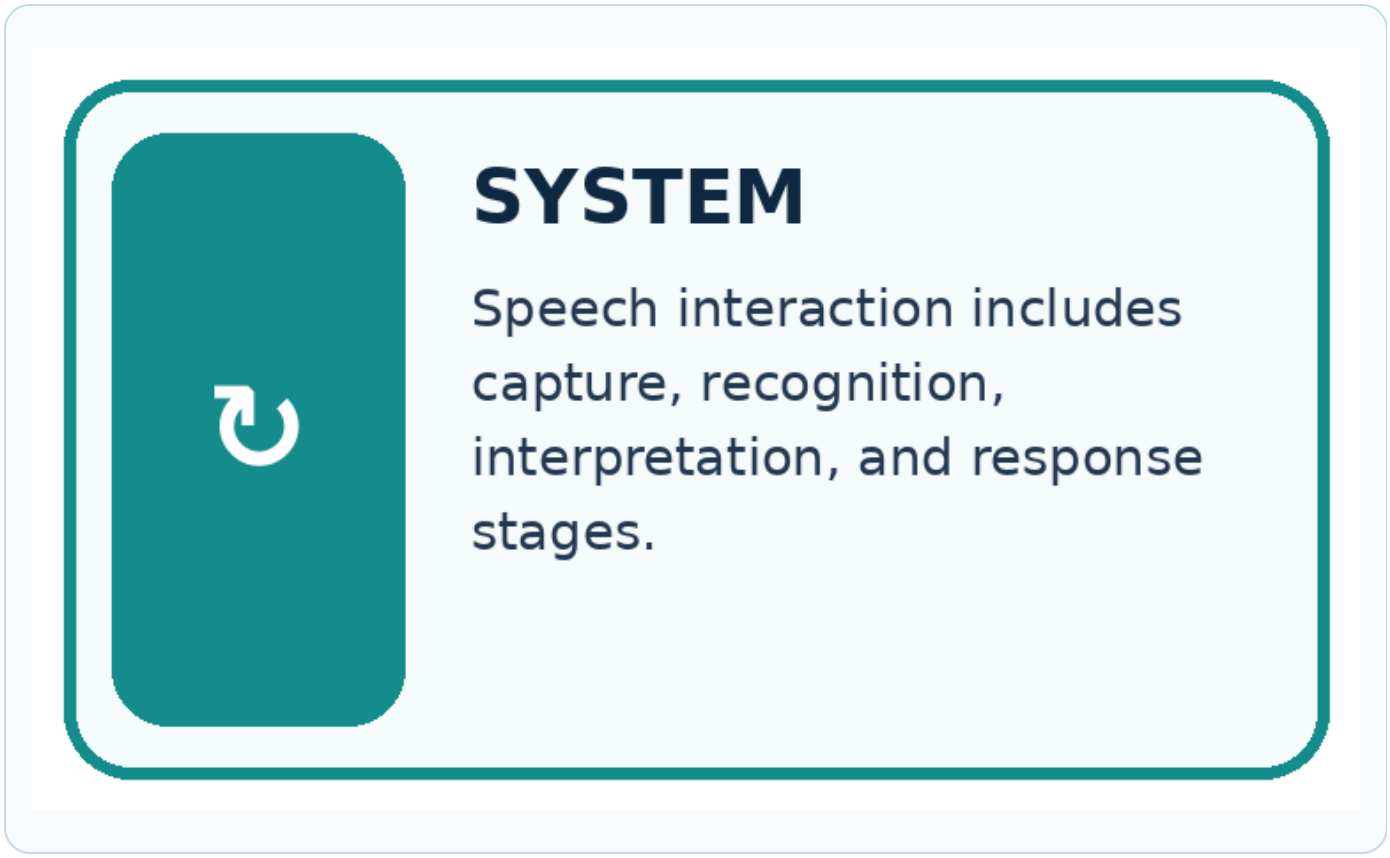
VISUAL MECHANISM

Input → Process → Output

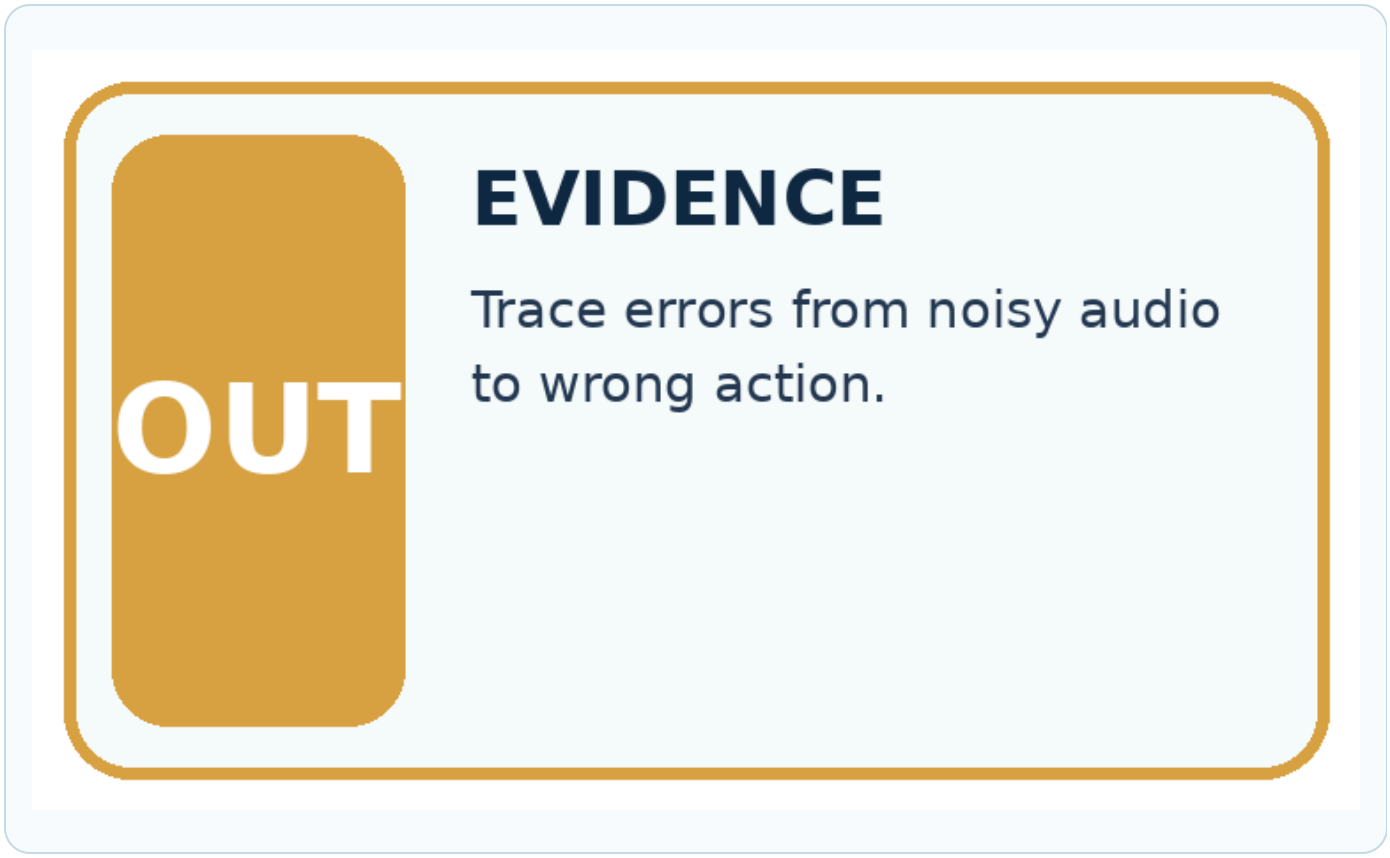
Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Speech interaction includes capture, recognition, interpretation, and response stages.

↓


OUT
EVIDENCE
Trace errors from noisy audio to wrong action.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately. Compare conflicting **evidence** without assuming one **modality** always wins.

Ask for clarification when the task needs unresolved information.

Speech pipelines and multimodal evidence — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.	Track source modality, segment/time references, transcription, interpretation and final claims separately. Compare conflicting evidence without assuming one modality always wins. Ask for clarification when the task needs unresolved information.	There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

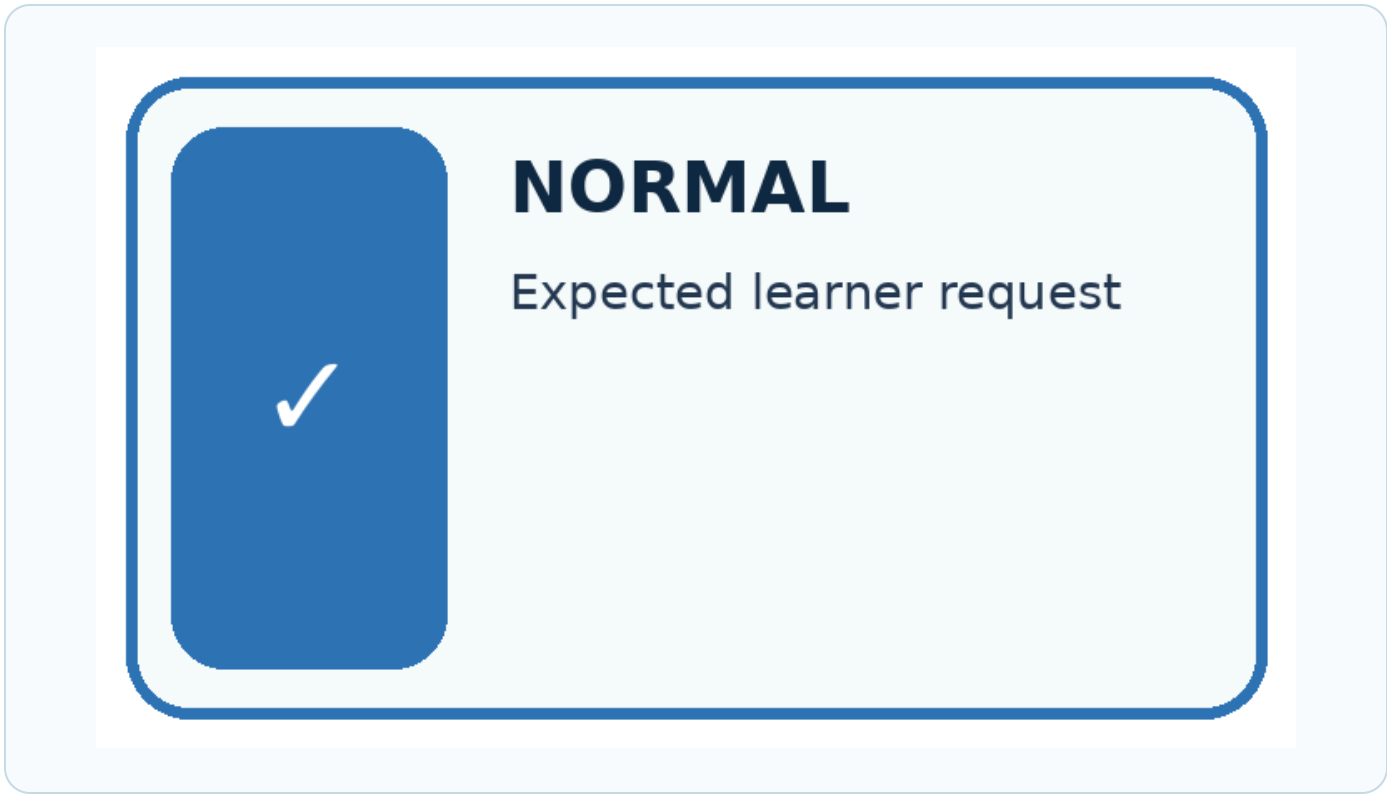
> Compare the supplied cases for Speech and Audio Systems. Identify the detail that changes the conclusion.

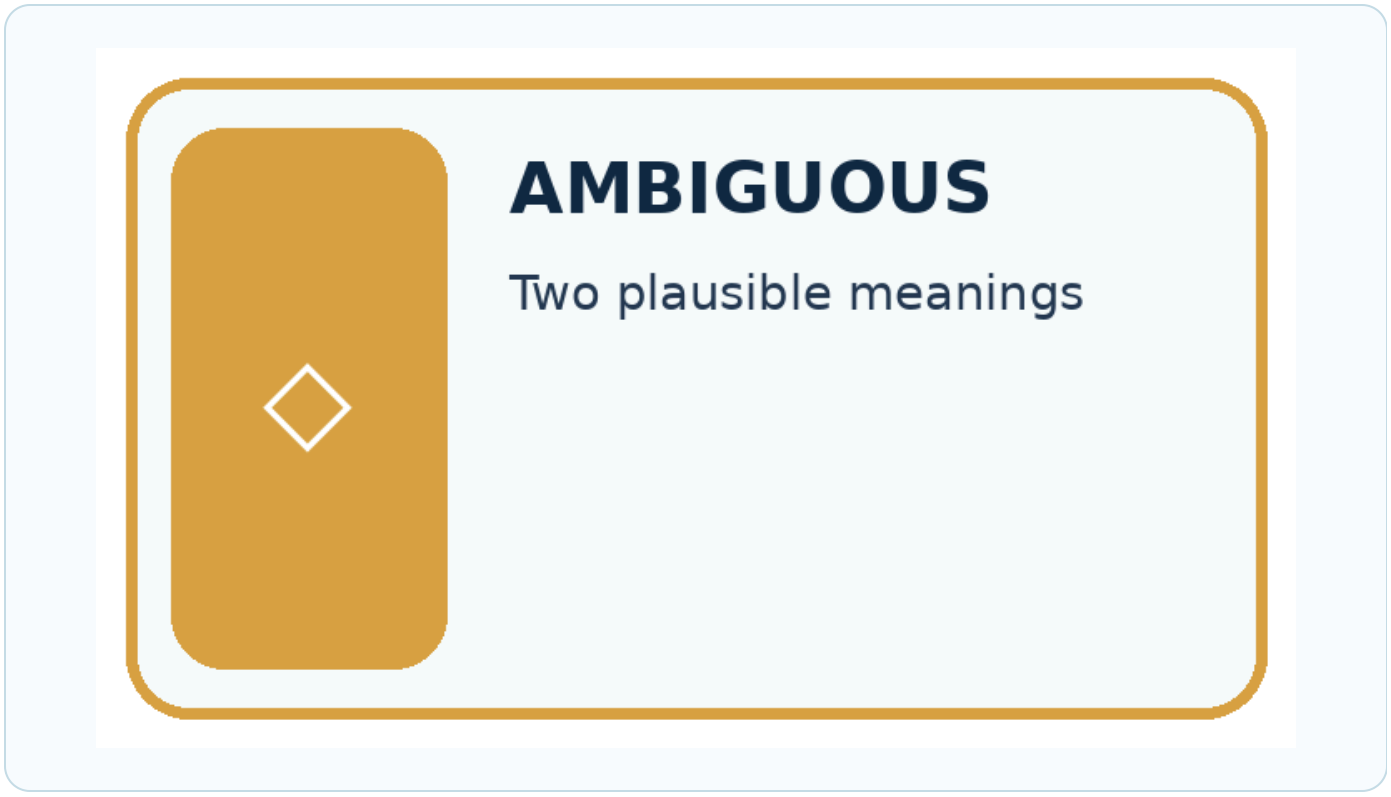
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Correct the transcript to “thirteen plus seven” with all later stages unchanged.

Predicted result

Prediction: The calculator and spoken result should become 20 under the declared pipeline.

Why the result follows

Reason: 13+7=20; the changed **input** removes the earlier recognition error.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

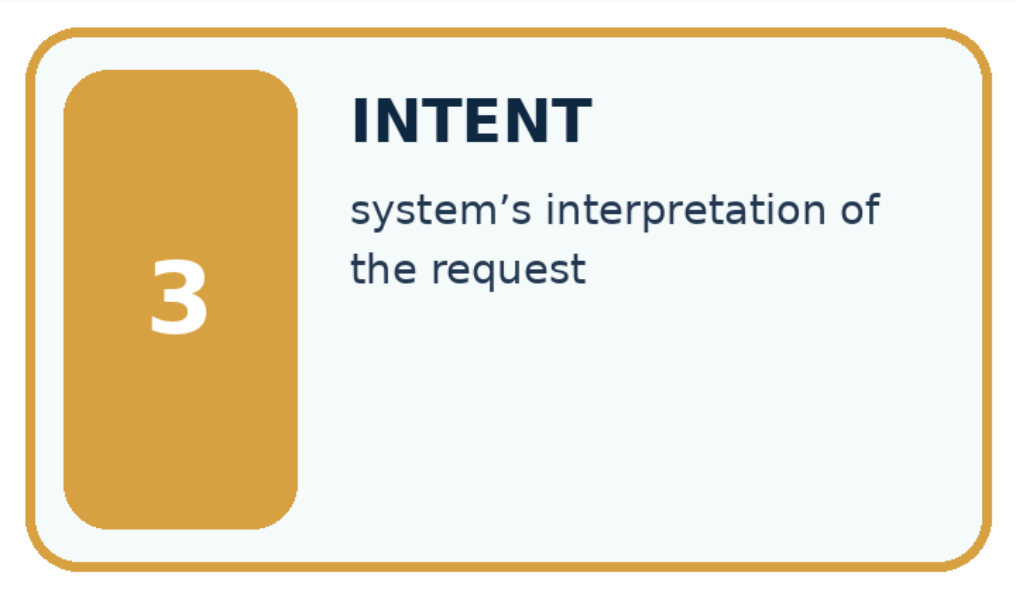
> Use the vocabulary of Speech and Audio Systems to explain a concrete decision, rather than only memorising definitions.

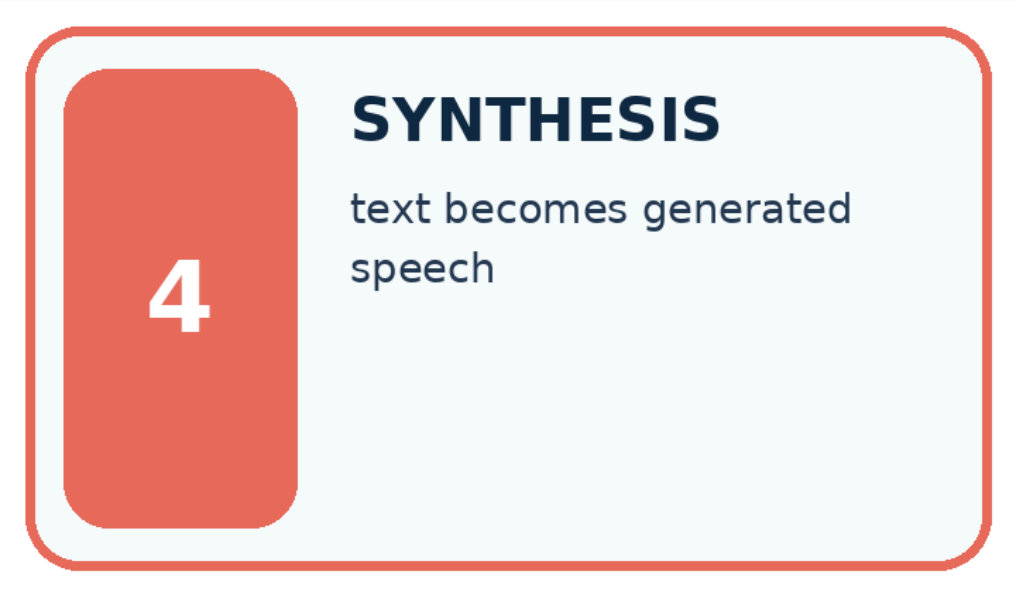
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
CAPTURE
microphone measures an audio signal
2
SPEECH RECOGNITI
signal becomes candidate words





Word	My own example	Not this / boundary
capture	Record a signal, observation or item of information for later use.	
speech recognition	Converting spoken audio into an estimated sequence of words.	
intent	The goal a user is trying to achieve, which may need clarification.	

Word	My own example	Not this / boundary
synthesis	Combining information into a new account or output while preserving its meaning and support.	

Words used in the closer explanation

Word	Meaning
Modality	A form of information such as audio, image or text
Transcription	A text representation of speech
Error propagation	An earlier error affecting later processing

---

STUDENT LESSON · 6 OF 18

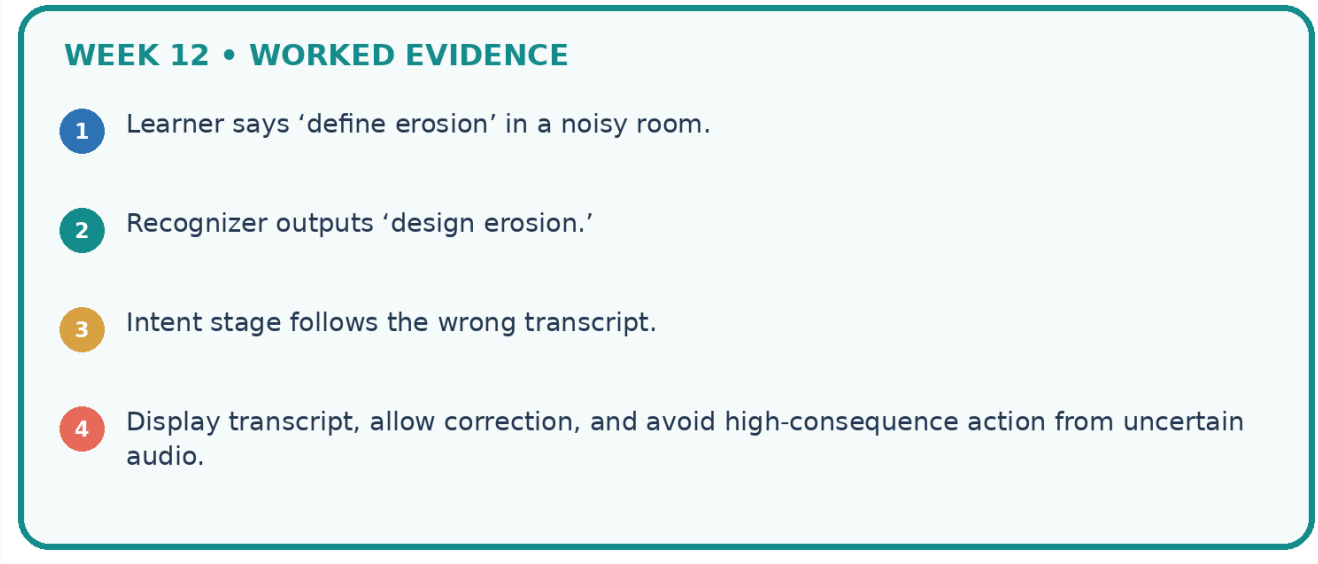
Follow a worked example

> Follow the worked trace for Speech and Audio Systems, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Learner says ‘define erosion’ in a noisy room.

2. Recognizer **output**s ‘design erosion.’

3. **Intent** stage follows the wrong transcript.

4. Display transcript, allow correction, and avoid high-consequence action from uncertain audio.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Worked explanation · Speech and Audio Systems

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Speech and Audio Systems

A fictional learner dictates a calculation to a voice assistant.

The facts we will use

Supplied facts

Known audio phrase: “thirteen plus seven”. Transcript: “thirty plus seven”. Calculator returns 37.

Supplied facts

Synthetic voice says “thirty-seven”. The example supplies ground truth; no real audio was analysed.

AI ACADEMY / AI-3/12

Speech and Audio Systems

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Recognise speech

Transcription changes thirteen into thirty.



2. Calculate text

30+7=37 is correct for the mistaken transcript.



3. Speak result

Speech synthesis repeats 37; fluent voice does not validate the input.



4. Confirm intended input

Read back the recognised expression; intended 13+7=20.

CHANGE ONE THING

Correct the transcript to "thirteen plus seven" with all later stages unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Recognise speech	Transcription changes thirteen into thirty.
2. Calculate text	30+7=37 is correct for the mistaken transcript.
3. Speak result	Speech synthesis repeats 37; fluent voice does not validate the input.
4. Confirm intended input	Read back the recognised expression; intended 13+7=20.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The first error occurs before arithmetic. Blaming the calculator would target a stage that correctly **process**ed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action.

Retain the original audio and transcript, where permitted, to separate recognition defects from later **process**ing errors.

What this example does not establish

Keep the stated limit

These supplied transcript facts do not establish how an actual recogniser performs for accents, noise or different speakers.

Keep the stated limit

Read the labelled picture: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?

Keep the stated limit

Change one thing: Correct the transcript to “thirteen plus seven” with all later stages unchanged. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Speech and Audio Systems claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A wrong spoken response means the language model caused the first error. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

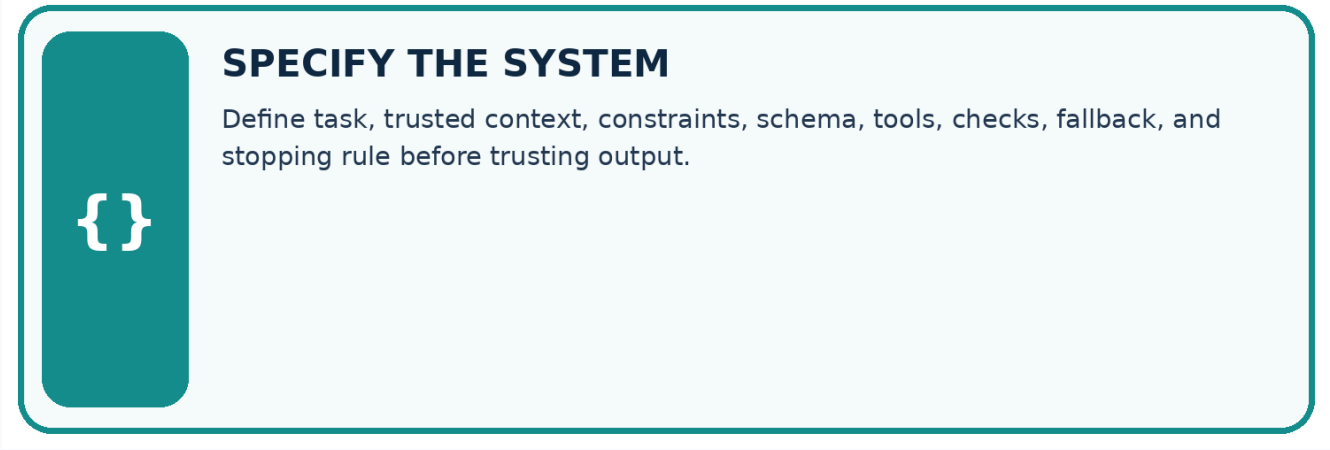
Practise together

> Practise Speech and Audio Systems with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Speech pipelines and multimodal **evidence**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Speech pipelines and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

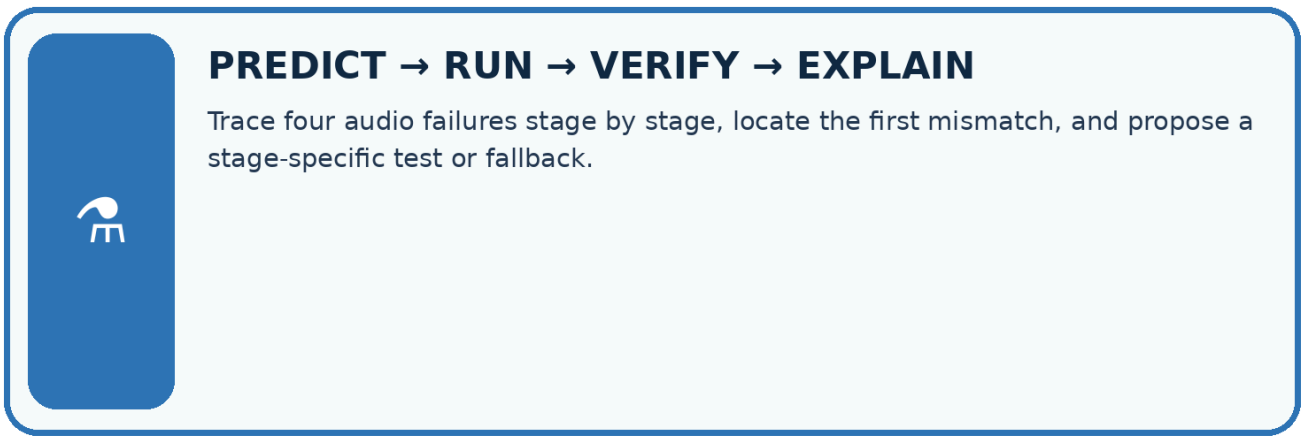
Run the investigation

> Investigate Speech and Audio Systems. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Trace four audio failures stage by stage, locate the first mismatch, and propose a stage-specific test or fallback.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

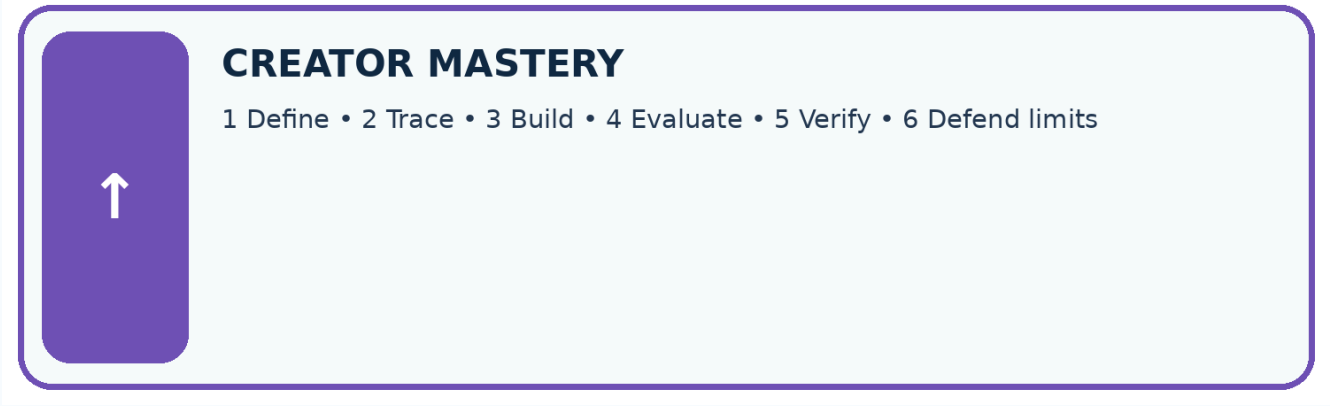
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Speech and Audio Systems. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

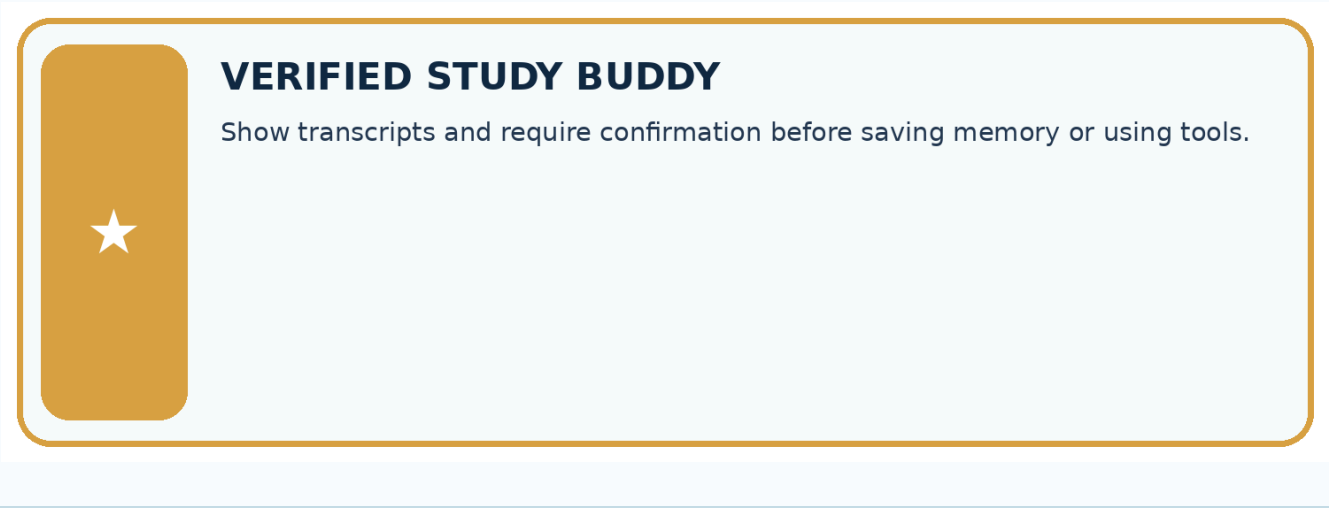
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Show transcripts and require confirmation before saving memory or using tools.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates.

STUDENT LESSON · 12 OF 18

Show what you know

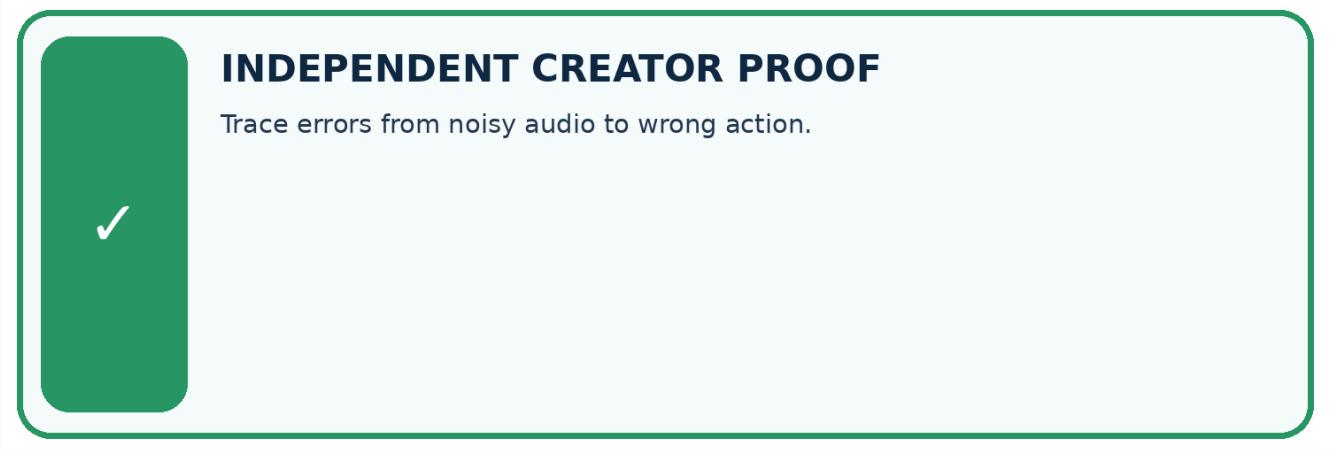
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Trace errors from noisy audio to wrong action.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Speech and Audio Systems case for supported practice. Repeated examples do not become fresh assessments.

Week 12 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A speech tool hears fifteen as fifty, then calculates correctly using fifty.

The question

> Where did the error enter?

Worked explanation

Transcription supplied the wrong number. Confirm the transcript or ask for clarification. Correct arithmetic cannot repair wrong **input**.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Trace a speech error through the system

Evidence or stage	Value or result
Speech	Fifteen
Transcript	Fifty, the first error
Calculator	Correct arithmetic using fifty
Result	Wrong for the original request

A speech tool hears fifteen as fifty, then calculates correctly using fifty.

> Where did the error enter?

Transcription supplied the wrong number. Confirm the transcript or ask for clarification. Correct arithmetic cannot repair wrong **input**.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

Known audio says “nineteen minus six”. **Transcription** says “ninety minus six”; the calculator returns84. Locate the first error and propose a check.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 12 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A recording says fifteen, but the transcript says fifty. The summary repeats fifty. Where did the error enter, and what should be checked first?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Show transcripts and require confirmation before saving memory or using tools.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> Audio says 'sixteen divided by four'; transcript says 'sixty divided by four'; calculator returns 15. Which stage first fails? What would a useful confirmation show?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Speech and Audio Systems, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional learner dictates a calculation to a voice assistant.

By the end, I can	How I will show it
Explain the mechanism and terminology of speech pipelines and multimodal evidence	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Speech and Audio Systems without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional learner dictates a calculation to a voice assistant.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Read back the recognised expression; intended 13+7=20.

1. Confirm intended **input**

2. Recognise speech

3. Calculate text

4. Speak result

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Confirm intended **input** Read back the recognised expression; intended 13+7=20. The first error occurs before arithmetic.

Blaming the calculator would target a stage that correctly **process**ed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action.

Retain the original audio and transcript, where permitted, to separate recognition defects from later **process**ing errors.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Correct the transcript to “thirteen plus seven” with all later stages unchanged. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The calculator and spoken result should become 20 under the declared pipeline.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The calculator and spoken result should become 20 under the declared pipeline. 13+7=20; the changed **input** removes the earlier recognition error.

Repair a missing building block

Review Visual Composition (ai-3/11). Goal: Explain the **mechanism** and terminology of image generation and provenance. Short practice: Which checks pass after that single change, and what still needs repair?.

Return to this week and explain what you changed.

Review Prompt Evaluation (ai-3/9). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Why is B's selection defensible here without calling it universally superior?.

Return to this week and explain what you changed.

Review Examples and Rubrics (ai-3/6). Goal: Write a prompt specification with observable response criteria. Short practice: Explain why neither original nor changed draft is a full pass, and provide a complete repaired card..

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

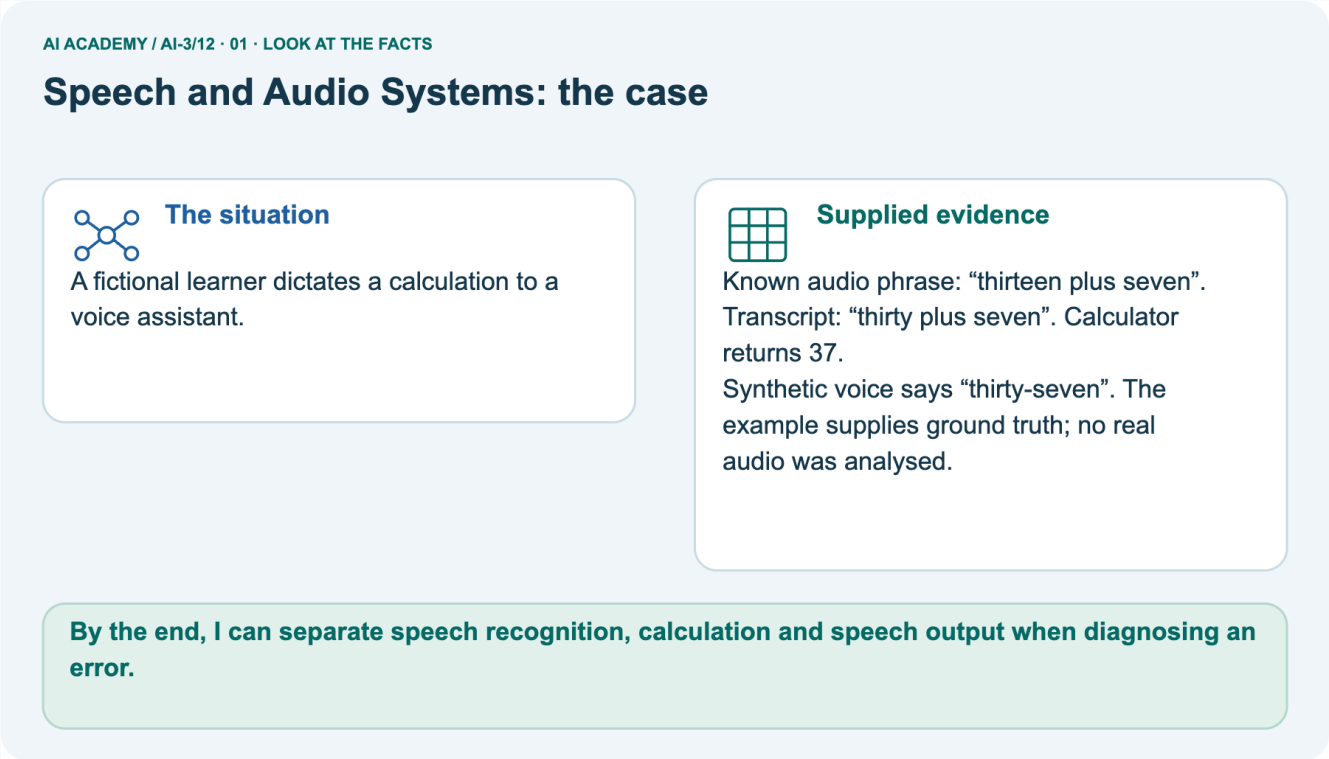
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Speech and Audio Systems graphic using its labels, deciding evidence and relevant limitation.

01. Speech and Audio Systems: the case



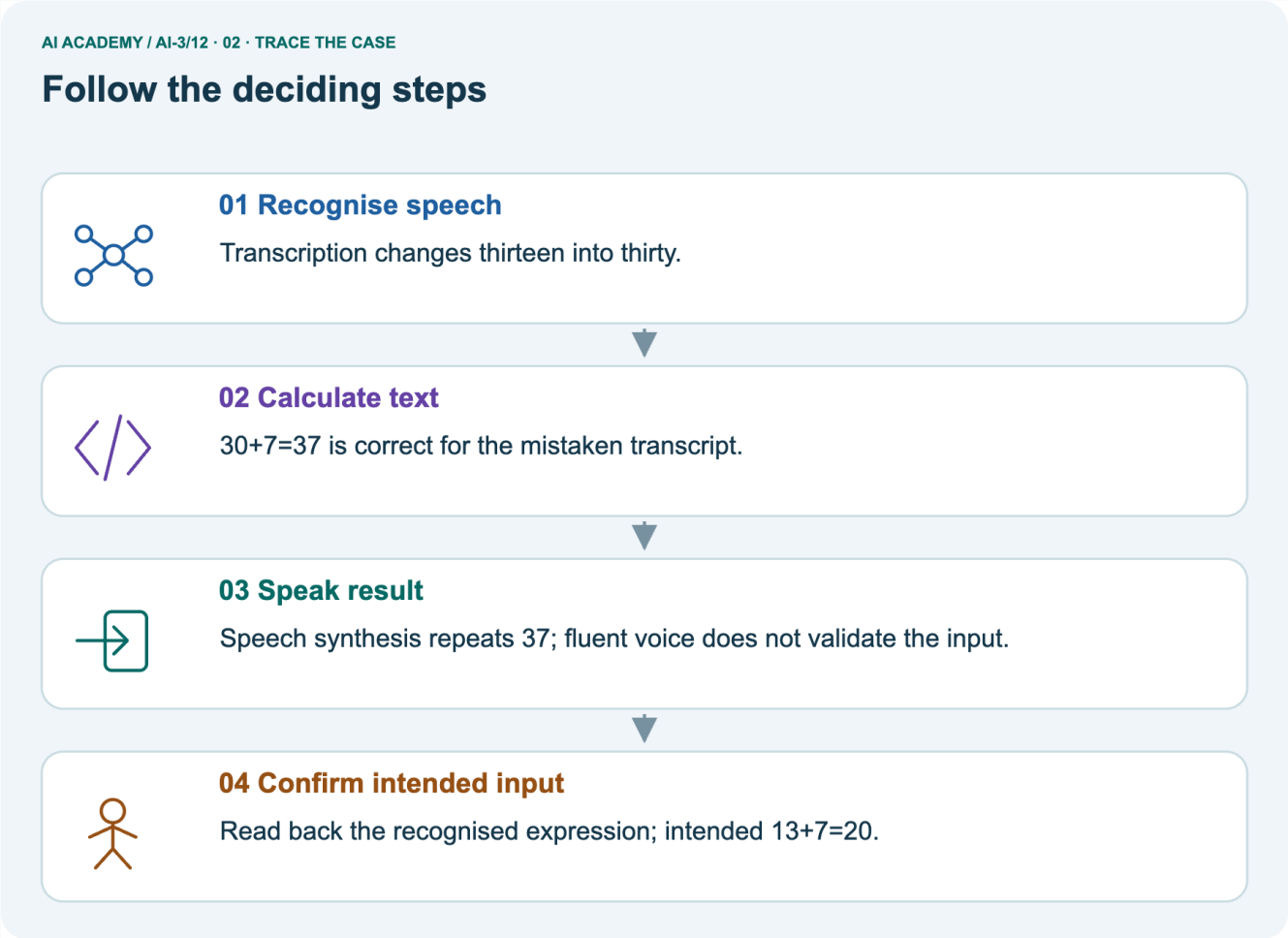
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional learner dictates a calculation to a voice assistant. Supplied facts: Known audio phrase: “thirteen plus seven”. Transcript: “thirty plus seven”. Calculator returns 37. Synthetic voice says “thirty-seven”.

The example supplies ground truth; no real audio was analysed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

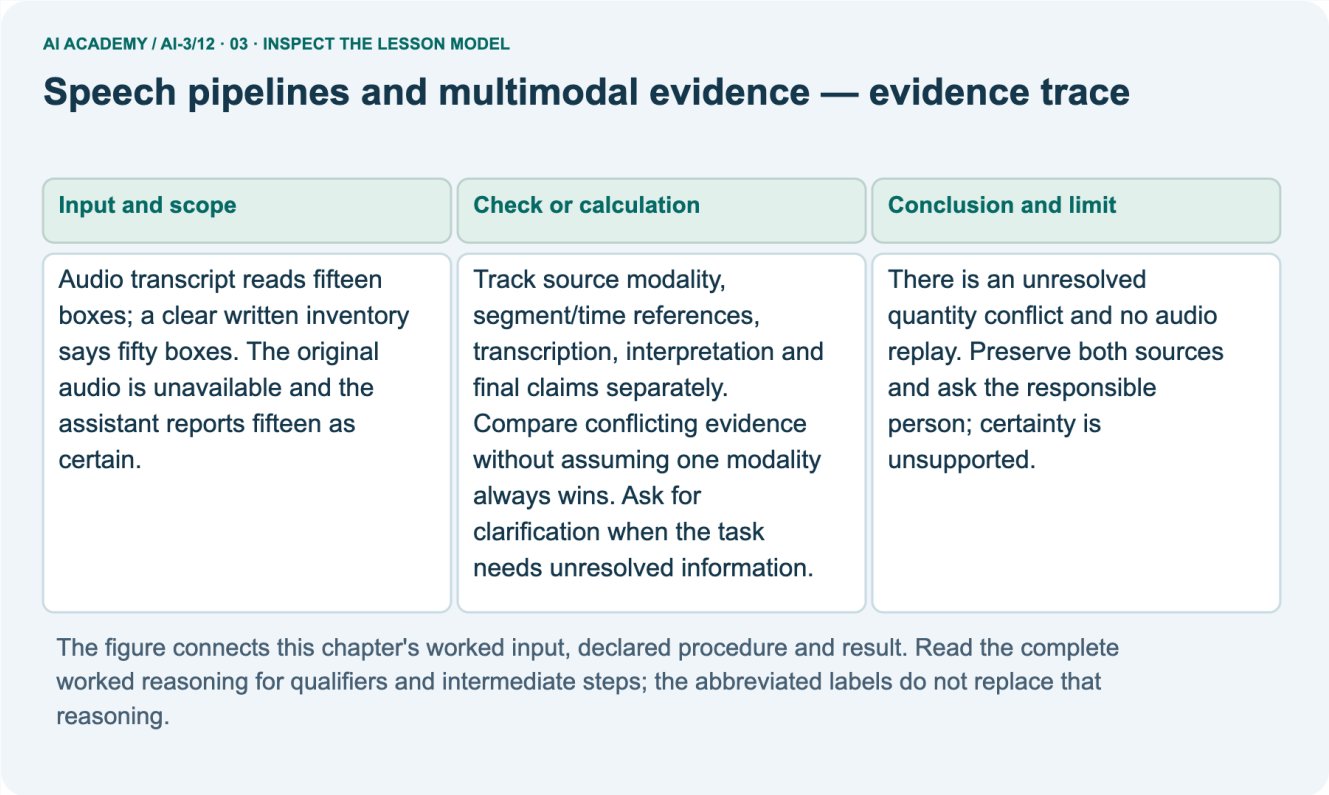


Follow the labelled picture with the matching explanation. Open the vector diagram

Recognise speech: **Transcription** changes thirteen into thirty. → Calculate text: 30+7=37 is correct for the mistaken transcript. → Speak result: Speech **synthesis** repeats 37; fluent voice does not validate the **input**. → Confirm intended **input**: Read back the recognised expression; intended 13+7=20.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Speech pipelines and multimodal evidence — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Speech pipelines and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

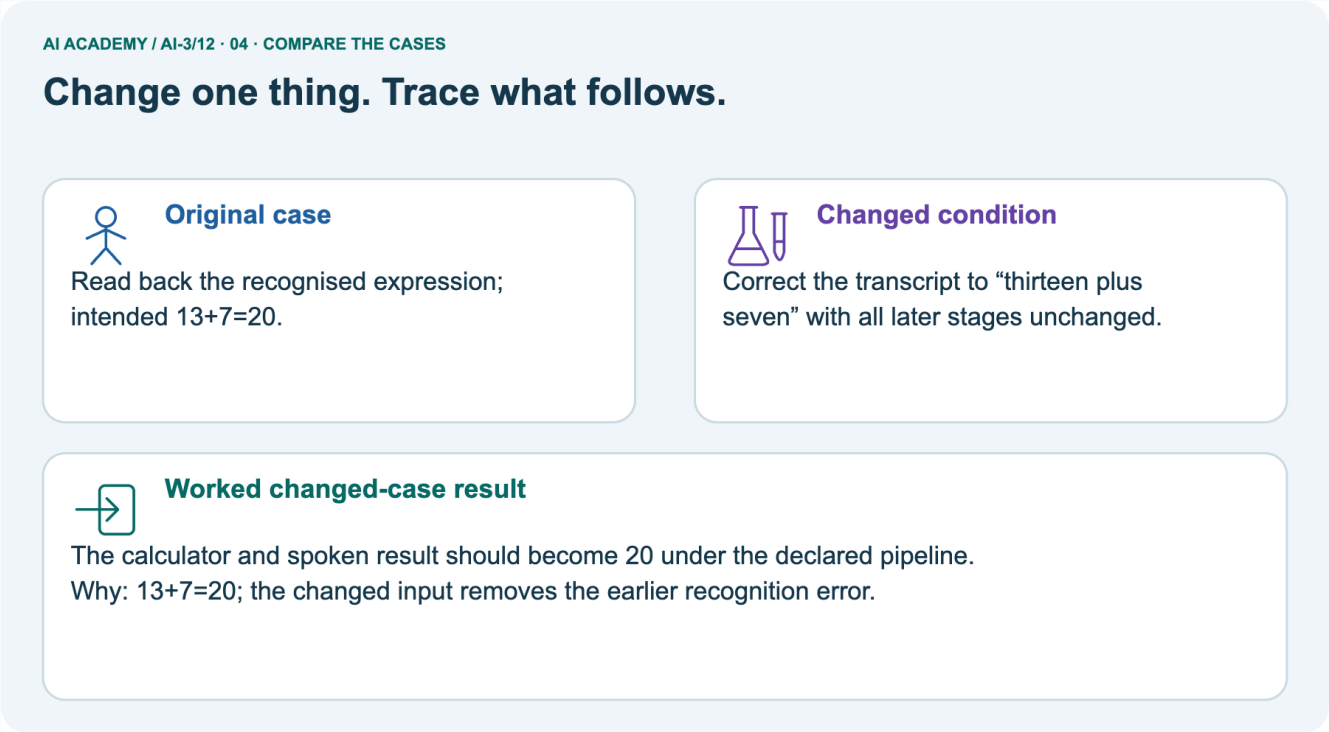
The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

Compare conflicting **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity conflict and no audio replay.

Preserve both sources and ask the responsible person; certainty is unsupported.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



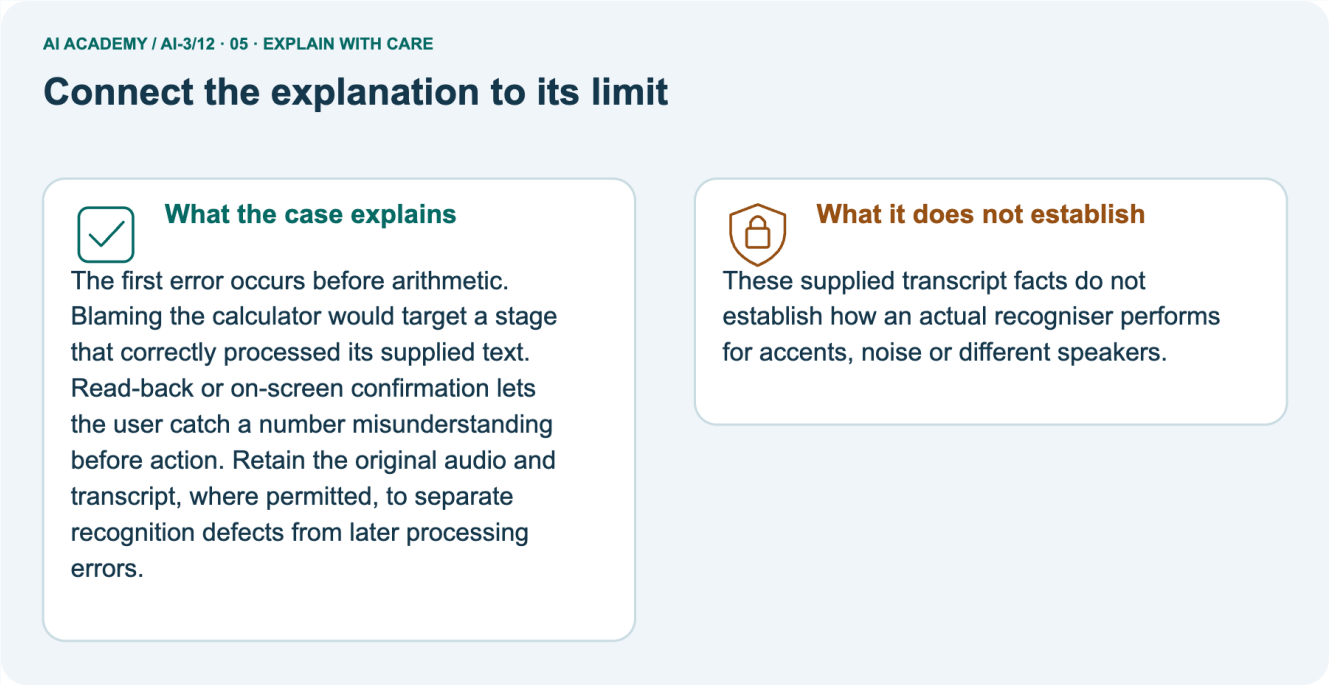
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Read back the recognised expression; intended 13+7=20. Changed condition: Correct the transcript to “thirteen plus seven” with all later stages unchanged.

Expected result: The calculator and spoken result should become 20 under the declared pipeline. Reason: 13+7=20; the changed **input** removes the earlier recognition error.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

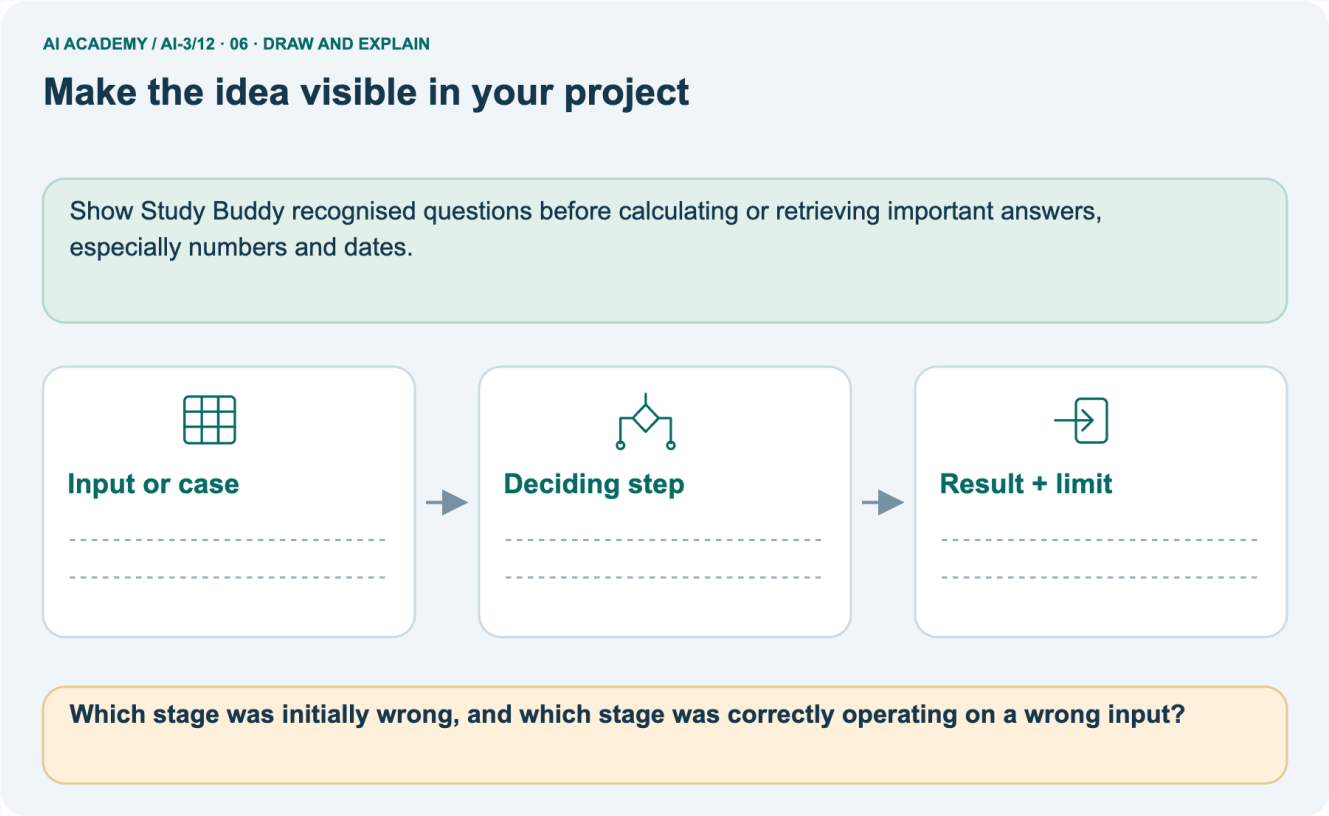
The first error occurs before arithmetic. Blaming the calculator would target a stage that correctly **process**ed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action.

Retain the original audio and transcript, where permitted, to separate recognition defects from later **process**ing errors.

Limit: These supplied transcript facts do not establish how an actual recogniser performs for accents, noise or different speakers.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates. Draw your **input**, deciding step, result and limit.

Supported question: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.